2 Introduction to Basic Graphs – Calculation-Based DI

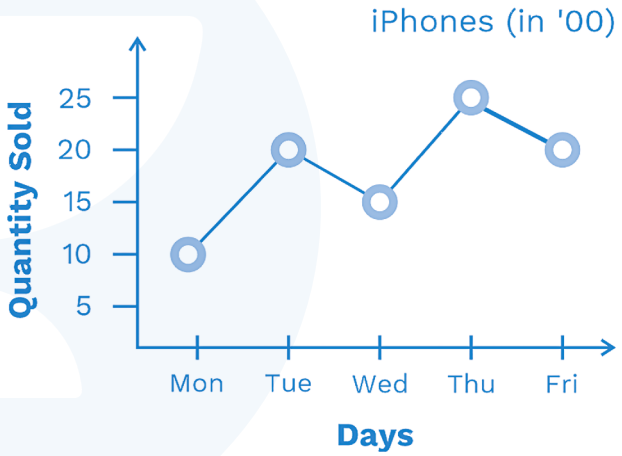
Introduction

- Calculation-based data interpretation (DI) sets appear regularly in CAT and other MBA entrance exams. In fact, exams like IIFT, XAT, NMAT, CET, etc., give more weightage to the calculation-based DI sets as compared to the reasoning-based DI sets.
- This module will help you understand the variety of charts (as part of the calculation-based DI) and the approach to solving them. We will start with the conventional data representation techniques and later on, cover the other form of charts.

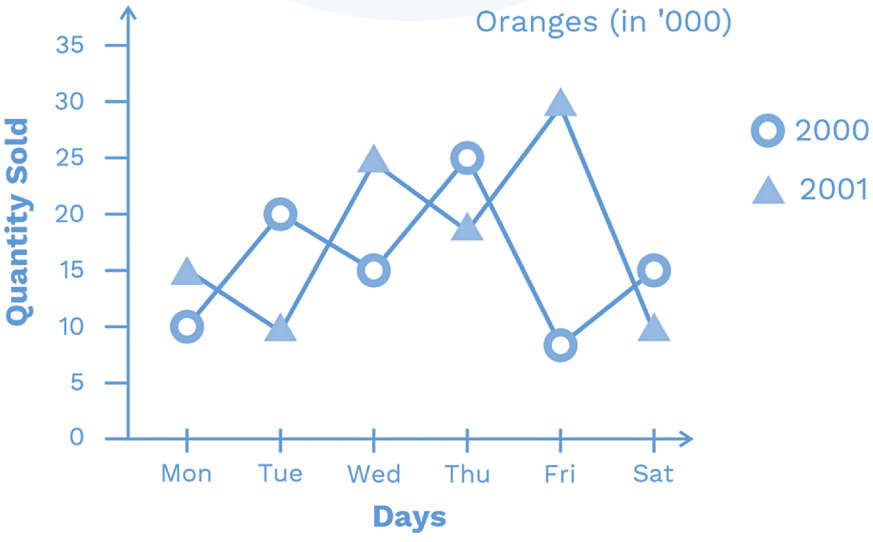
Plenty of solved examples are provided to help you understand the concept in detail. After going through the concepts, you can solve the questions from the three practice exercises that have been provided (in the increasing order of difficulty) along with the detailed solution to each question.

- Graphical representations of information are very useful as it helps the presenter communicate data in a way that's easy for the viewer to interpret and draw conclusions. There are various forms of data representation. The prominent among them are:
  - Line graph
  - Bar graph
  - Pie chart
  - Tabular form
  - Radar chart, scatter diagram, etc.
- In fact, any geometrical figure can be used to represent the data, depending upon the situation. (For example, you should not be surprised if you see a triangular or hexagonal chart in your exam). So, let's understand each form of data representation with suitable examples.

Line Graph



(Single Line Graph)



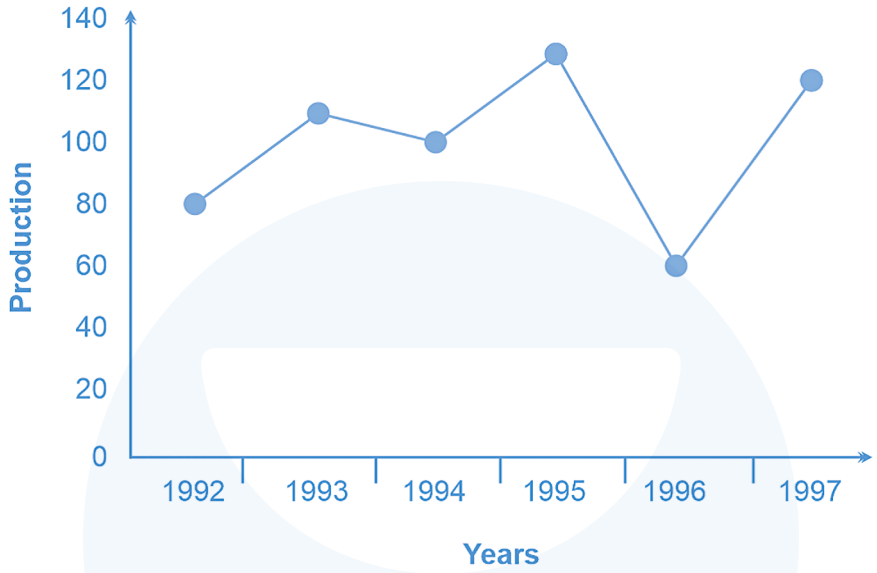
(Double Line Graph)

- A line graph represents the data or set of different data values on a two-dimensional plane by using a straight line that changes the direction according to the given value.
- It is widely used to represent trends over a given period of time, e.g., share index, growth in revenue, speed, etc.

Example 1:

Directions for Questions 1 to 4: Analyse the graph carefully and answer the questions that follow.

The below-given graph represents annual food grain production (in million tonnes) on the *y*-axis and the years from 1992 to 1997 on the *x*-axis.



1. Find the approximate percentage decrease in food grain production from 1993 to 1994.
(A) 87.5%
(B) 37.5%
(C) 9.09%
(D) 11.11%

Solution: (C)

Production in 1993 = 110 million tonnes

Production in 1994 = 100 million tonnes

Required percentage decrease

$$= \left(\frac{110-100}{110}\right) \times 100\%$$

$$= \frac{10}{110} \times 100\% = \frac{100}{11}\% = 9.09\%$$

Hence, option (C).

2. In which year was the average production for the entire period approximately equal to the average production of 1994 and 1996?
(A) 1992
(B) 1993
(C) 1994
(D) 1995

Solution: (A)

Production in 1994 = 100 million tonnes

Production in 1996 = 60 million tonnes

Average production of 1994 and 1996

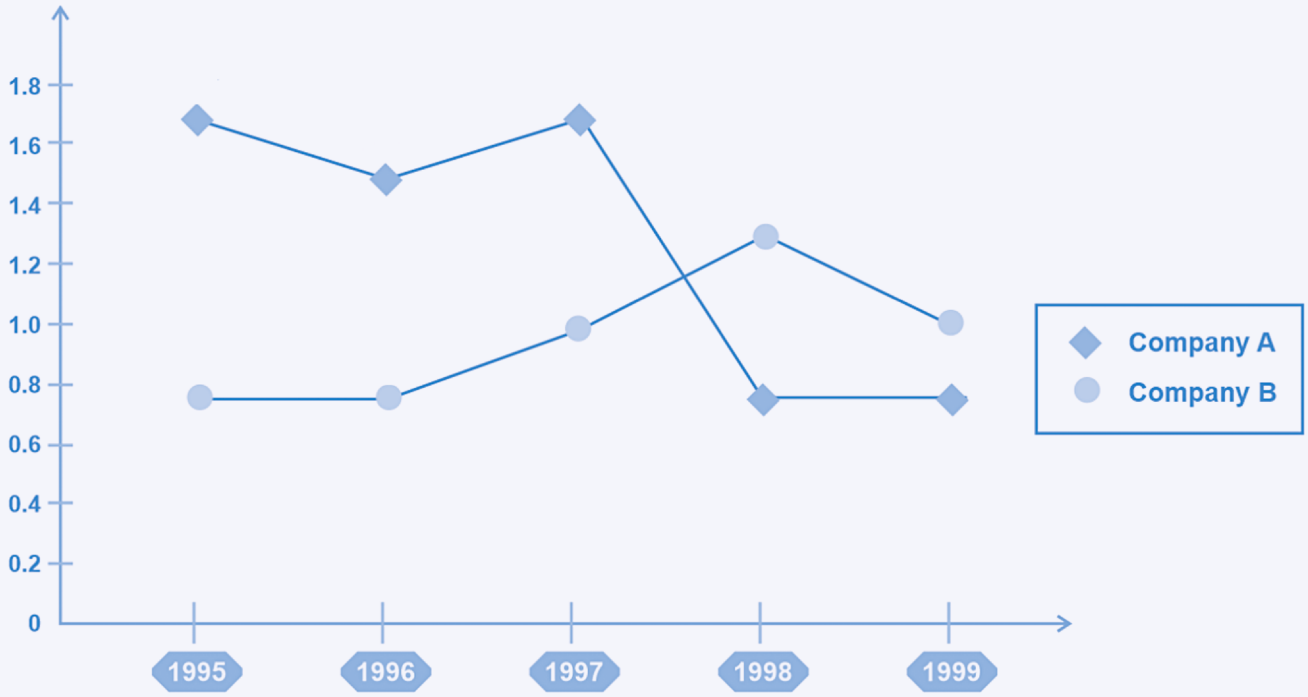
$$= \frac{(100+60)}{2} = 80 \text{ million tonnes}$$

Hence, average production of 1994 and 1996 is equal to the production in the year 1992, i.e., 80 million tonnes.

Hence, option (A).

Rack Your Brain

The line graph below shows the ratio of exports and imports of two companies for the period 1995–1999.



If the exports of company A were twice that of company B in 1995 and the total imports of company A during the years 1995–1999 was 20 million dollars, then find the approximate total imports of company B.

3. For how many years, the production in that year was more than the average production over the given period?

Solution: 3

Average production from 1992 to 1997

$$= \frac{(80+110+100+130+60+120)}{6} = \frac{600}{6} = 100$$

Production in the years 1993, 1995, and 1997 was more than the average production over the years.

Hence, the answer is 3.

4. Assuming that the food grain production in 1992 is 60% more than that in 1991, then what was the production in 1991?

Solution: 50

Production in 1992 = 80

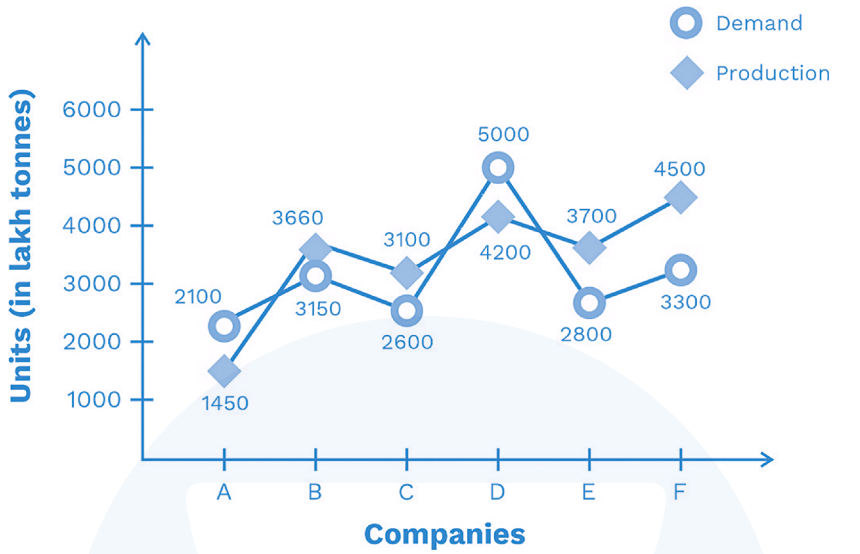
Thus, production in 1991 $= \frac{80}{160} \times 100 = 50$

Hence, the answer is 50.

Example 2:

Directions for Questions 5 to 8: Study the following graph carefully and answer the questions that follow.

Demand and production of different companies for a particular year



5. What is the ratio of the number of companies having more demand than production to those having more production than demand.

(A) 2:3
(B) 1:2
(C) 2:1
(D) 3:2

Solution: (B)

Companies having more demand than the production are A and D.

Companies having more production than demand are B, C, E, and F.

So, the required ratio $= \frac{2}{4} = 1:2$.

Hence, option (B).

Tips by Toppers

Easy questions help you assess your accuracy and speed.

Try to maintain 100% accuracy from the very beginning of your preparations.

6. What is the difference between the average demand and the average production of all the companies (in lakh tonnes)? (Round off your answer to the closest integer).

Solution: 277

Average demand

$$= \frac{(2,100 + 3,150 + 2,600 + 5,000 + 2,800 + 3,300)}{6}$$

$$= \frac{18,950}{6} = 3,158.33$$

Average production

$$= \frac{(1,450 + 3,660 + 3,100 + 4,200 + 3,700 + 4,500)}{6}$$

$$= \frac{20,610}{6} = 3,435$$

So, the required difference $= 3,435 - 3,158.33 = 276.67 \approx 277$

Hence, the answer is 277.

7. The production of company A is approximately what percentage of the demand of company C?

(A) 55%
(B) 60%
(C) 65%
(D) 50%

Solution: (A)

The required percentage $= \frac{1,450}{2,600} \times 100\% \approx 55\%$

Hence, option (A).

Note: 50% of 2,600 = 1,300

5% of 2,600 = 130

8. For which company is the production as a percentage of the demand the highest?

(A) B
(B) C
(C) E
(D) F

Solution: (D)

In such questions, you really don't need to calculate the exact percentage unless the values turn out to be very close. You can actually do the comparison in fraction form, as shown below:

For company B, $\frac{\text{Production}}{\text{Demand}} = \frac{3,660}{3,150} = \frac{366}{315} = \frac{37}{32}$

For company C, $\frac{\text{Production}}{\text{Demand}} = \frac{3,100}{2,600} = \frac{31}{26}$

For company E, $\frac{\text{Production}}{\text{Demand}} = \frac{3,700}{2,800} = \frac{37}{28}$

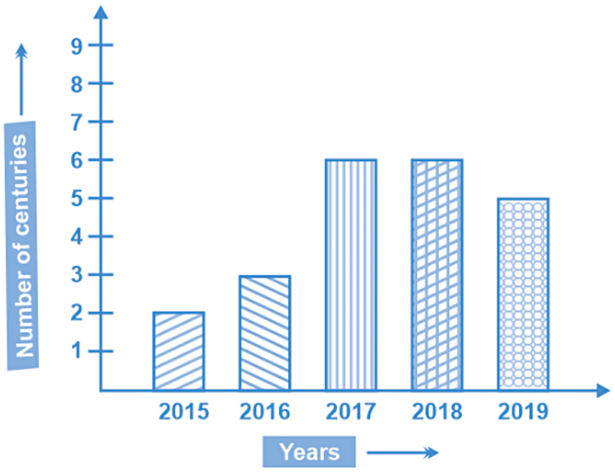
For company F, $\frac{\text{Production}}{\text{Demand}} = \frac{4,500}{3,300} = \frac{45}{33}$

Clearly, $\frac{45}{33}$ is the highest fraction.

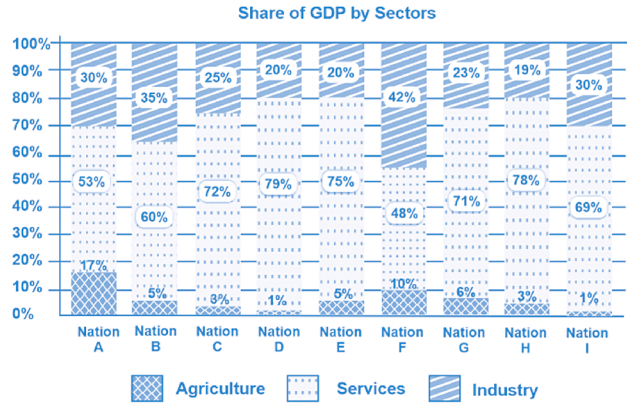
So, the answer is company F.

Hence, option (D).

Bar Graph



(Normal Bar Graph)



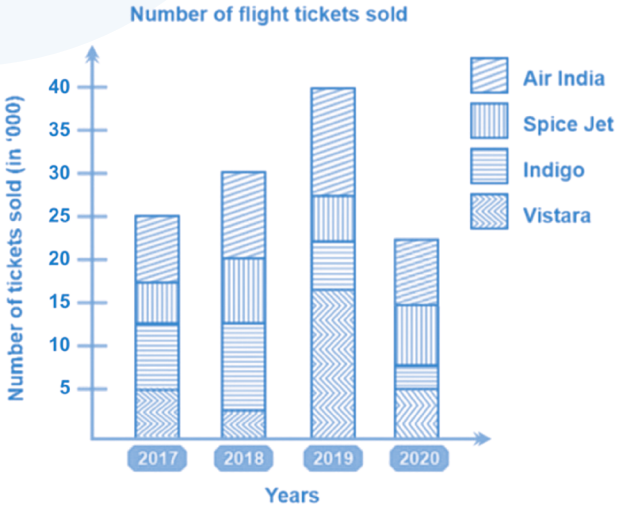
(Stacked Bar Graph)

If a set of data elements is represented on a coordinate plane using two axes, with the help of rectangularly placed bars, the length of which indicates the numerical value, quantity or the measure of data elements, it is called a bar graph.

A stacked bar graph (or stacked bar chart) is a chart that uses bars to show comparisons between categories of data, but with the ability to break down and compare parts of a whole. Each bar in the chart represents a whole, and segments in the bar represent different parts or categories of that whole.

Example 3:

Directions for Questions 9 and 10: Study the following graph carefully and answer the questions that follow.



9. What is the ratio of the number of tickets sold by Vistara and Spice Jet together in 2018 to the number of tickets sold by Vistara and Air India in 2020?

(A) $\frac{2}{3}$

(B) $\frac{5}{4}$

(C) $\frac{4}{5}$

(D) $\frac{3}{2}$

Solution: (C)

Number of tickets sold by Vistara in 2018 = 2,500.

Number of tickets sold by Spice Jet in 2018 = 7,500.

So, the number of tickets sold by Vistara and Spice Jet in 2018 = 10,000.

Number of tickets sold by Vistara in 2020 = 5,000.

Number of tickets sold by Air India in 2020 = 7,500.

So, number of tickets sold by Vistara and Air India in 2020 = 12,500.

So, the required ratio $= \frac{10,000}{12,500} = \frac{4}{5}$

Hence, option (C).

10. What is the difference between the total number of tickets sold by Air India and Indigo from 2017 to 2020?

(A) 7,500
(B) 7,000
(C) 8,000
(D) 12,500

Solution: (D)

In such questions, instead of calculating the total tickets sold by each airline and taking the difference, it is better to calculate the difference for each year and then add the difference of values to get the desired answer.

(This is called the cumulative difference method. Just make sure to consider the sign when you calculate A – B).

For 2017, tickets by Air India – tickets by Indigo = 7,500 – 7,500 = 0.

For 2018, tickets by Air India – tickets by Indigo = 10,000 – 10,000 = 0.

For 2019, tickets by Air India – tickets by Indigo = 12,500 – 5,000 = 7,500.

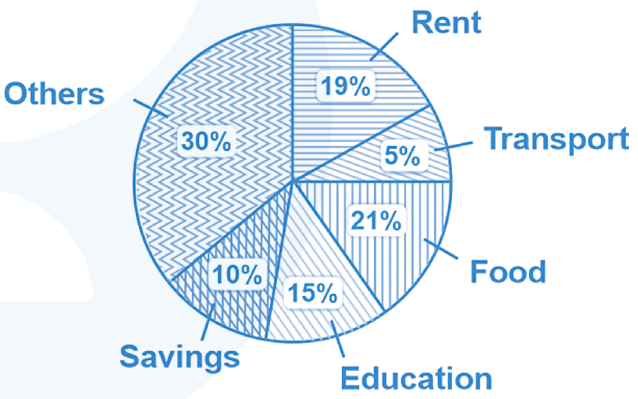
For 2020, tickets by Air India – tickets by Indigo = 7,500 – 2,500 = 5,000.

So, the difference between the total tickets sold by the two airlines over the period = 0 + 0 + 7,500 + 5,000 = 12,500.

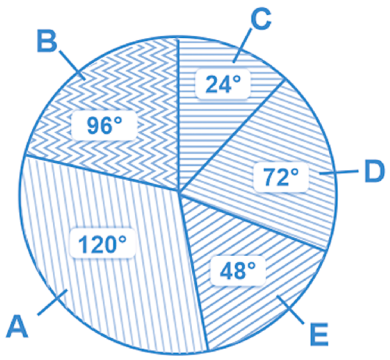
Hence, option (D).

Note: In the cumulative difference method shown above, if the tickets sold by Air India are less than that by Indigo, then the (A – B) value would have been considered negative.

Pie Chart



Percentage Pie Chart



Degree Pie Chart

A circular statistical graph which is divided into slices to illustrate numerical proportion is known as pie-chart.

Pie charts can be categorised into three types based on the value.

1. Absolute value pie-chart
2. Percentage value pie-chart
3. Degree value pie-chart

Points to Remember

- In a percentage distribution pie-chart, the total percentage is always equal to 100%.
- In a degree distribution pie-chart, the sum of the total central angle is always equal to 360°.
- Also, remember the following % and equivalent degree conversion values for quick calculations.

%	Degree
100%	360°
50%	180°
25%	90°
10%	36°
5%	18°
1%	3.6°

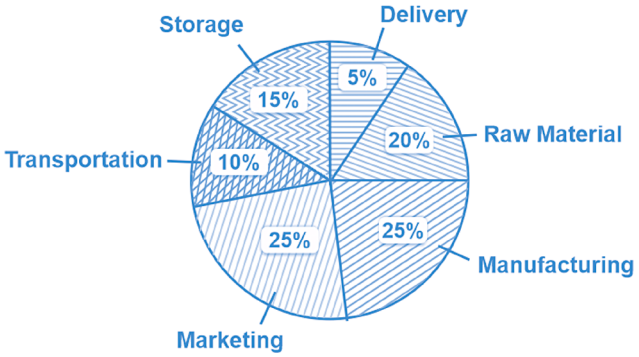
Rack Your Brain

What is the equivalent percentage for 111.6°?

Example 4:

Directions for Questions 11 and 12: Study the following chart carefully and answer the questions that follow.

The pie-chart below shows the various costs incurred (in percentage) by sports shoe manufacturing company Tuma. Answer the following based on the pie-chart.



11. If 10,000 pairs of shoes are produced and the cost incurred on delivery is ₹5 lakhs, what should be the selling price of a pair of shoes so that the company will earn a profit of 20% overall?

(A) ₹1,500
(B) ₹1,000
(C) ₹1,200
(D) Cannot be determined

Solution: (C)

From the pie-chart given in the question, it can be understood that the cost incurred on delivery is 5% of the total cost incurred. It is also known that the cost incurred for delivery is ₹5 lakhs.

If 5% amounts to ₹5 lakhs, the total cost incurred will be ₹100 lakhs = ₹1 crore.

So, the cost of one pair of shoes

$$= \frac{100 \text{ lakh}}{10{,}000} = ₹1{,}000$$

So, the SP of one pair of shoes should be ₹1,200 for 20% profit.

Hence, option (C).

12. The price of a pair of shoes is marked ₹2,500 which is 25% more than its cost price. What is the cost of raw material for a single pair of shoes?

Solution: 400

CP of a single pair of shoes

$$= \frac{100 \times 2500}{125} = ₹2{,}000$$

So, the cost of raw material for a single pair of the shoes

$$= \frac{20}{100} \times 2{,}000 = ₹400$$

Hence, the answer is 400.

Tabular Form

	Male	Female
< 20 Years	20	30
20 to 30 Years	35	45
> 30 Years	45	25

This is the simplest form of data presentation. It helps to represent even a large amount of data in an engaging, easy to read, and coordinated manner. The data are arranged in rows and columns. Many a time, the situation arises where there are so many parameters attached to the data. In such a situation, the tabular form of data representation comes very handy.

Rack Your Brain

Can you represent the following data in a tabular form?

There is a college of 1,000 students with three disciplines—Commerce, Science, and Arts. One-fourth of the total students is in Science, two-thirds of the rest are in Commerce, and the remaining are in Arts. In each discipline, the students are divided into three divisions—first year, second year, and third year. The number of students in these three divisions is 20%, 30%, and 50%, respectively, for each of the disciplines. In each division, the number of males and females is in the ratio of 2:3.

Example 5:

Directions for Questions 13 to 16: Study the following chart carefully and answer the questions that follow.

For making an alloy, different elements are used. The price inflation index of these elements is given in the table below, considering the year 2009 as the base year. For example, if the unit price of element C was ₹100 in 2009, it rose to ₹105 in 2013. Based on this information, answer the following questions. (Assume the cost of each alloy as ₹100 in 2009, unless mentioned otherwise).

Elements	2009	2010	2011	2012	2013
C	100	102	101.8	103.4	105
Li	100	99.8	97.8	95.6	96
S	100	101.3	100.4	100.7	101
W	100	103	107	106	110
G	100	101.4	103	104.7	107
P	100	98.5	98	95.5	97
La	100	103	106.6	110	114

17. Which element's price rose by the maximum percentage in the last year over the previous year?

(A) La
(B) W
(C) G
(D) P

Solution: (B)

Let's look at the fractional change in the prices of each element from 2012 to 2013.

$$La = \frac{4}{110} \qquad G = \frac{2.3}{104.7}$$

$$W = \frac{4}{106} \qquad P = \frac{1.5}{95.5}$$

Here, $\frac{4}{106} > \frac{4}{110}$ (denominator of the former is smaller)

$$\frac{2.3}{104.7} < \frac{2.3+1.3}{104.7+1.3} = \frac{3.6}{106}$$

$$\therefore \frac{2.3}{104.7} < \frac{4}{106}$$

So, W had the highest percentage rise over the last year.

Hence, option (B).

18. If the total cost of each element used in making an alloy was equal in 2009, and the cost incurred making an alloy was 1 crore in 2009, how much would the same alloy cost in 2013?

(A) ₹1,04,28,571
(B) ₹1,06,23,086
(C) ₹1,09,23,973
(D) ₹1,03,34,267

Solution: (A)

Since the prices of each element was the same (₹100) in the year 2009, and we are also given that the total cost of each element used in the alloy was also equal, we can say that the weights of all elements used were also equal.

Also, the average price for all the alloys in the year 2009 was ₹100. So, let's check the average price in 2013, and that will help us check the increase in the total cost of the alloy too.

Average of prices in 2013:

Average of prices

$$= \frac{105+96+101+110+107+97+114}{7} = 104.28571$$

$$\text{So, new total cost} = ₹1 \text{ crore} \times \frac{104.28571}{100}$$

$$= 1,04,28,571$$

Hence, option (A).

19. If the total cost of each element used during making an alloy was equal in 2009, what is the annualised rate of inflation for making an alloy from 2009 to 2013?

(A) 1.06%
(B) 1.17%
(C) 1.28%
(D) Cannot be determined

Solution: (B)

As calculated earlier, the new average cost is 104.286 in 2013 and it was ₹100 in 2009.

Let r be the rate of inflation per year.

$$\therefore (1+r)^4 = \frac{104.286}{100} = 1.04286 \text{ (using CAGR formula)}$$

From the final value, r seems to be fairly small and hence we can approximate the expansion to $1 + 4r = 1.04286$.

So, approx. $r = 1.07\%$

So, the closest value is 1.06%.

Hence, option (B).

20. If the average yearly rate of inflation was 1.2% for the 4 years, for how many elements prices rose at a higher rate than the average rate over the 4 years?

Solution: 4

If yearly rate of inflation was 1.2%, the cost in 2013 would be = cost $100 \times (1 + 1.2\%)^4$

≈ 104.8.

As can be seen in the table, four elements have cost more than 104.8 in 2013.

So, four elements rose by more than that over the 4 years.

Hence, the answer is 4.

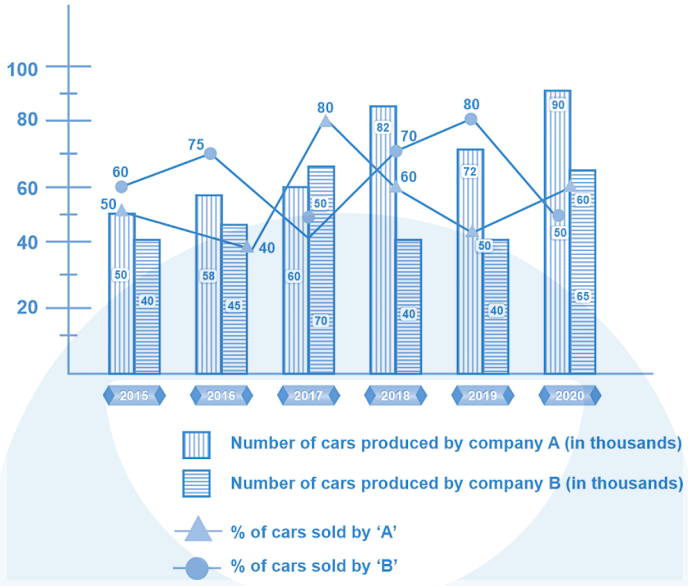
Practice Exercise

Level of Difficulty – 1

Set 1

Directions for Questions 1 to 5: Study the following information carefully and answer the given questions.

The following graph shows the number of cars produced and the percentage of produced cars sold by two companies 'A' and 'B' from 2015 to 2020.



1. What is the total number of cars produced by company A which remained unsold in all six years together?
(A) 1,42,600
(B) 1,52,600
(C) 1,76,600
(D) 2,00,000

2. What is the ratio of the number of cars produced by company B in 2019 to the number of cars that remained unsold by company A in the year 2015?
(A) 8:5
(B) 4:3
(C) 5:4
(D) 4:1

3. What is the difference between the total number of sold cars and unsold cars of company B in the years 2015, 2017, and 2020?

4. The number of cars sold by 'A' in 2016 is what percent of the number of cars sold by 'B' in the year 2020?
(A) 82.5%
(B) 69.17%
(C) 71.38%
(D) 74.33%

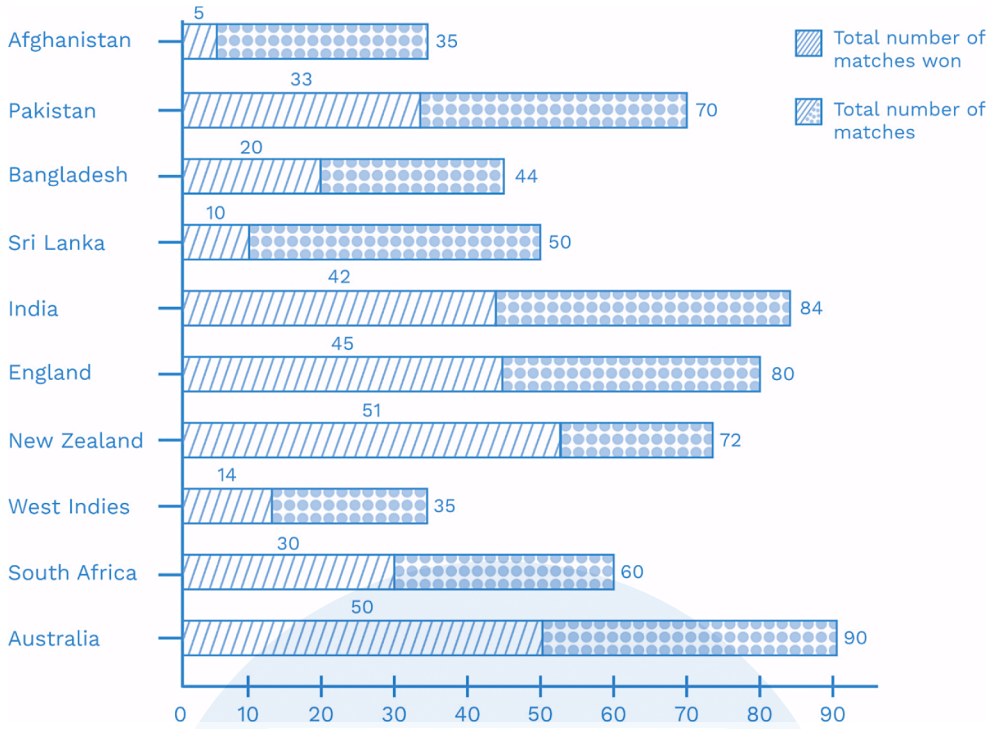
5. What is the difference in values of the number of cars sold in 2017 and the number of cars sold in 2019 by company A?

Set 2

Directions for Questions 6 to 10: Refer to the cumulative bar chart given below and answer the questions that follow.

Several matches were played and won by any team during 2010–2020.

Note: No matches resulted in a tie.



6. What is the ratio of the number of matches lost by team India and the number of matches lost by New Zealand?

(A) $\frac{1}{2}$

(B) $\frac{3}{1}$

(C) $\frac{2}{1}$

(D) $\frac{1}{3}$

7. What is the difference between the winning percentage of Sri Lanka to the winning percentage of South Africa?

(A) 30%
(B) 20%
(C) 25%
(D) 40%

8. How many teams have more than or equal to 50% winning ratio?

9. Which of the following team has the highest percentage of wins?

(A) South Africa
(B) New Zealand
(C) India
(D) Australia

10. What is the losing percentage of the West Indies?

(A) 50%
(B) 55%
(C) 60%
(D) 65%

Set 3

Directions for Questions 11 to 15:

The following table gives some incomplete information on the marks obtained by four students Paul, Bruno, Marcus, and Edinson in four different subjects.

	Maths	Science	English	French	Total
Paul	80	70	-	80	-
Bruno	60	-	60	-	-
Marcus	-	-	60	-	220
Edinson	90	-	70	60	-

Further, it is known

1. Paul's average marks were 72.5 in all four subjects.
2. The average marks in maths for all four students were 67.5.
3. Bruno's average marks for all four subjects were 1.5 times of Marcus' marks in Maths.
4. Bruno and Edinson scored the same marks in French. The average French marks were the same as the average maths marks.
5. The total marks in science were 10 less than the total marks in maths for all four students.

11. What is the sum of marks scored by Paul in English and Marcus in Maths?
(A) 80
(B) 90
(C) 100
(D) Cannot be determined

12. Who scored the lowest total?
(A) Paul
(B) Bruno
(C) Marcus
(D) Edinson

13. Who is the highest scorer in science?
(A) Paul
(B) Bruno
(C) Marcus
(D) Edinson

14. Who is the lowest scorer in maths?
(A) Paul
(B) Bruno
(C) Marcus
(D) Edinson

15. Who is the highest scorer in French?
(A) Paul
(B) Bruno
(C) Marcus
(D) Edinson

Level of Difficulty – 2

Set 1

Directions for Questions 1 to 5

The line graph given below shows the sales (in 100 units) of watches of different watch manufacturing companies viz, Casio, Titan, Maxima, and Noise in a small country, in various years. It is also given that,

Profit = Revenue – Expenditure

Revenue = Sales (by volume) × Selling price of each watch

$$\text{Profit \%} = \frac{\text{Profit}}{\text{Expenditure}} \times 100$$

Also, the market share of the company (MS) is defined as the volume of the sales of the company as a percentage of the total sales volume of all the given companies.

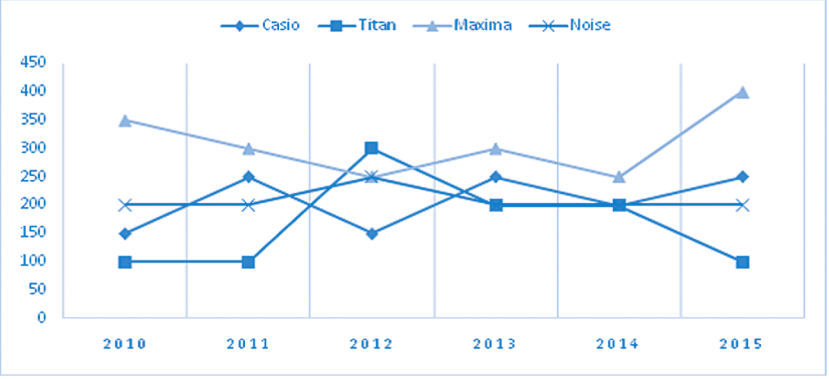
1. In the year 2014, if the profit percent on selling each Casio watch was 25% and the selling price of each Casio watch was ₹1,000, then what was the expenditure incurred by Casio in making watches?
(A) 1.5 crores
(B) 1.6 crores
(C) 2.4 crores
(D) 2.5 crores

2. The market share of the 'Maxima' company is highest in which year?
(A) 2015
(B) 2013
(C) 2011
(D) 2010

3. In the year 2012, the expenditure of Casio, Titan, Maxima, and Noise are in the ratio of 5:2:7:8. Which of the company had the highest profit percentage in 2013?
(A) Casio
(B) Titan
(C) Noise
(D) Cannot be determined

4. Which watch company had the highest growth rate in sales (volume-wise) during the period 2010 to 2014?
(A) Casio
(B) Titan
(C) Noise
(D) Maxima

5. Which company has the highest revenue in the year 2016, if sales volume of Casio, Titan, Maxima, and Noise, have grown by 30%, 20%, 10%, and 30%, respectively, as compared to the previous year, and the prices of their watches (in the same order) in the year 2016 were in the ratio 2:4:1:3?
(A) Maxima
(B) Casio
(C) Titan
(D) Noise

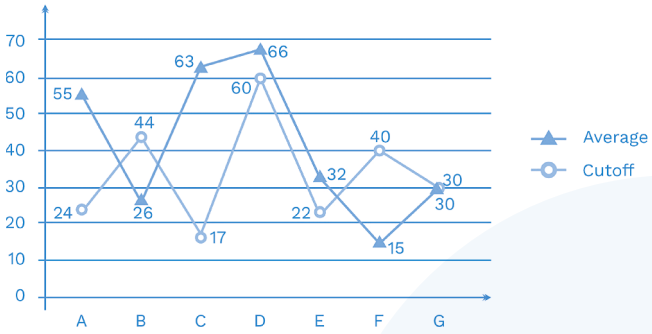


Set 2

Directions for Questions 6 to 10

The following graph shows the variation of average marks of seven different batches of seven students each (from different categories), and the cutoff marks in the admission test of XLRD College.

It is also known that no candidate scored above 70 in the test.



6. At most, how many students can pass in batch F?

7. For how many batches at the most, all seven students can qualify?

8. At most how many students (in total) can pass from all the seven batches?

9. If the performance of a batch is determined based on the number of students clearing the cutoffs in that batch, which is the best performing batch if the number of students scoring 70 marks is maximum possible?

(A) Batch A
(B) Batch C
(C) Batch D
(D) Either batch A or batch C

10. At most how many students can pass in batch G?

Set 3

Directions for Questions 11 to 15

The breakup of the animals in XYZ national park by category major is given in the pie-chart. The horizontal bar graph shows the number of animals who are from the insect category and are imported from different national parks.

Note: All the animals are imported from the given five national parks.

Chart – 1: Breakup of the animals

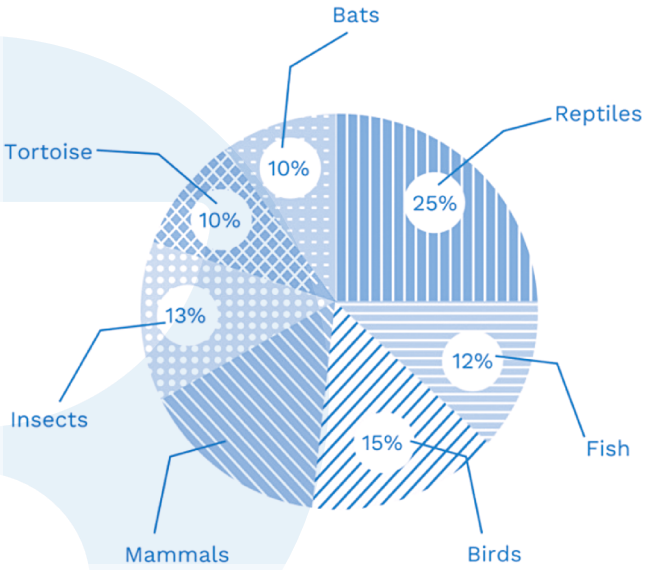
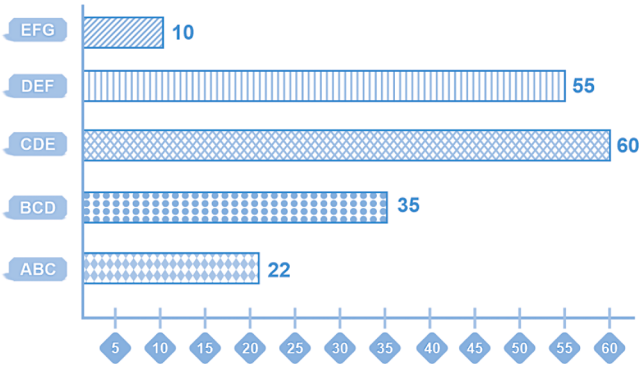


Chart – 2: Number of insects imported from different national parks



11. Find the total number of animals from birds category in all the national parks.
(A) 200
(B) 210
(C) 220
(D) 230

12. 8% of all animals are from DEF national park. What is the largest possible percentage of bats that can be from DEF national park rounded off to the nearer integer?
(A) 33%
(B) 25%
(C) 56%
(D) 41%

13. If the total number of mammals from BCD, CDE, and DEF national parks are the average of the number of animals from the insect category from BCD, CDE, and DEF national park together. Then, find the number of mammals from ABC and EFG together.
(A) 144
(B) 160
(C) 200
(D) 232

14. Find the total number of animals in XYZ national park.
(A) 1,000
(B) 1,400
(C) 1,290
(D) 1,420

15. Find the difference between the number of reptiles to the number of insects in XYZ national park.
(A) 233
(B) 155
(C) 67
(D) 168

Level of Difficulty – 3

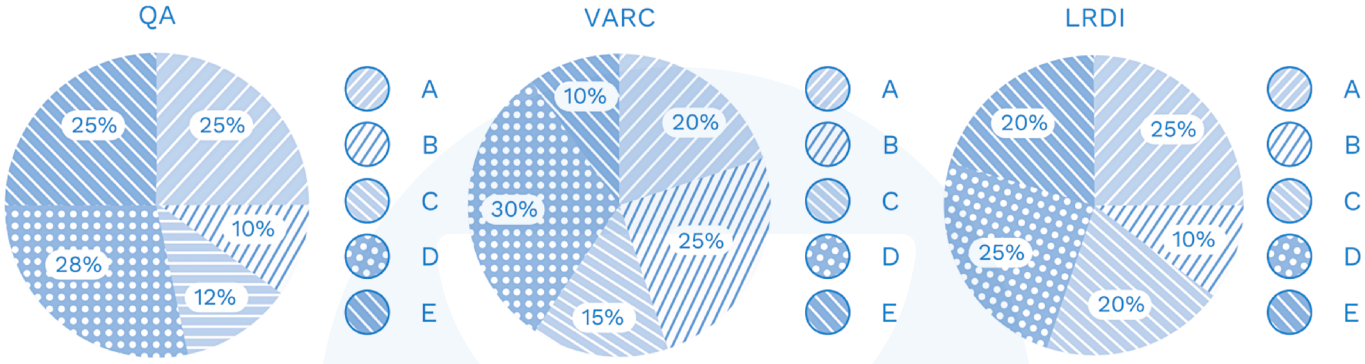
Set 1

Directions for Questions 1 to 5

Five students namely A, B, C, D, and E gave an entrance examination that included three sections namely quantitative aptitude (QA), verbal ability and reading comprehension (VARC), and logical reasoning and data interpretation (LRDI).

The first pie-chart given below provides the marks obtained by each student in the QA section as a percentage of the total marks obtained by five students in the QA section.

The second and third pie chart provides similar percentage breakups for the VARC section and LRDI section, respectively. It is also known that the total marks obtained were the same for exactly three of the five students.



Total marks obtained = Marks in QA section + Marks in VARC section + Marks in LRDI section

1. For how many students, the marks obtained in the LRDI section as a percentage of his/her total marks obtained is greater than 39%?
(A) 0
(B) 3
(C) 2
(D) Cannot be determined

2. Find the ratio of the marks obtained by D in the VARC section and the marks obtained by A in the QA section.
(A) 61:25
(B) 71:20
(C) 67:30
(D) 78:25

3. If the marks obtained by C in the VARC section are 292.5, then find the marks obtained by D in the QA section.
(A) 240
(B) 220
(C) 210
(D) 180

4. If the marks obtained by E in the LRDI section are 120, then find the average of the total marks obtained by B.
(A) 82.5
(B) 84
(C) 87.5
(D) 81.25

5. What is the difference between total marks obtained by C and E?

Direction for Questions 6 to 10

The bar chart below shows the number of eggs bought by the shopkeeper of a bakery store for a particular month from 11th to 18th. The line chart shows the percentage consumption of eggs as a percentage of total eggs available in stock that day.

Total eggs in stock during a day = Number of eggs bought by the shopkeeper on that day + Number of eggs carried over from the previous day.

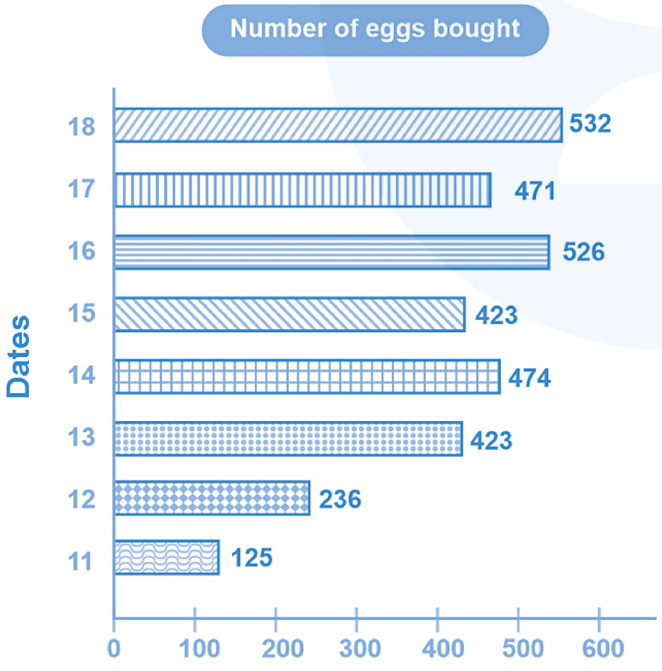
Note: 1. Consider only the integer part of the eggs consumed. (For example, if 28.5 eggs are consumed, then consider it as 28).

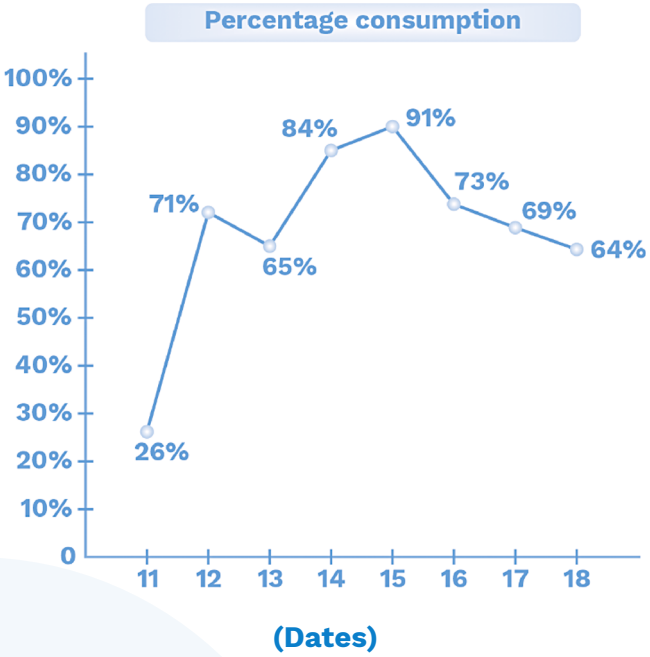
2. There were no eggs carried over on 11th.

For example, on 11th out of 125 eggs, 26% is consumed = 26% of 125 = 32.5 = 32.

Total eggs on 12th = 236 + (125 – 32) = 329.

Number of eggs consumed on 12th = 71% of 329 = 233.59 = 233, and so on.





6. Which day witnessed the highest number of eggs available in stock?
(A) 13th
(B) 16th
(C) 18th
(D) 14th

7. How many days witnessed a decrease in the total number of eggs available in stock as compared to the previous day?
(A) 2
(B) 3
(C) 1
(D) 4

8. Which day witnessed the highest consumption of eggs?
(A) 13th
(B) 17th
(C) 15th
(D) 14th

9. What was the percentage change in consumption of eggs on the 14th over the previous day?
(A) 58.3%
(B) 63.5%
(C) 71.2%
(D) 47.23%

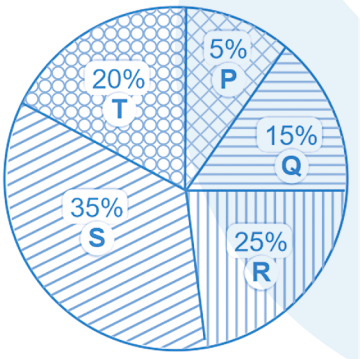
10. What is the absolute difference between the total number of eggs available at the store on the 18th and the 13th?
(A) 160
(B) 170
(C) 208
(D) 252

Direction for Questions 11 to 15

Zyntra is an Indian online store which sells Bluetooth earphones from only five companies, i.e., P, Q, R, S, and T.

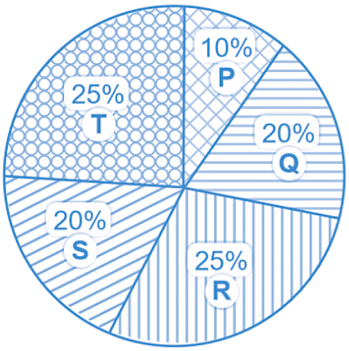
Pie chart I shows the percentage breakup of the total sales (in terms of the number of units sold) of Bluetooth earphones on Zyntra from these five companies, and pie chart II gives the percentage breakup of the total Bluetooth earphones sold by each of these five companies in India, in January 2020. (It is also known that these companies sell their Bluetooth earphones only in India).

Pie-chart I



Total = 400 (in thousand)

Pie-chart II



Total = k (in thousand)

(where 'K' is an unknown value)

11. What should be the minimum value of K (in thousand)?
(A) 300
(B) 650
(C) 900
(D) 700

12. Which company has sold the minimum percentage of its total Bluetooth earphones on Zyntra?
(A) T
(B) Q
(C) P
(D) Cannot be determined

13. If at least 25% of the total Bluetooth earphones sold by Q and not more than 50% of the total Bluetooth earphones sold by R, in January 2020 in India, were on Zyntra, what can be the maximum value of K (in thousand)?
(A) 1,000
(B) 1,200
(C) 900
(D) 600

14. What can be the maximum number of companies for which the percentages of their Bluetooth earphones sold on Zyntra is greater than or equal to 50% of their total sales in India, in January 2020?
(A) 0
(B) 1
(C) 2
(D) None of these

15. What is the absolute difference (in thousands) between the Bluetooth earphones sold by companies S and T in India on Zyntra?
(A) 40
(B) 50
(C) 60
(D) 70

Solutions

Level of Difficulty – 1

1. (C)

The given information can be tabulated as shown below:

	Number of Cars Produced	Number of Cars Sold	Number of Cars Unsold
2015	50,000	50,000 × 50% = 25,000	25,000
2016	58,000	58,000 × 40% = 23,200	34,800
2017	60,000	60,000 × 80% = 48,000	12,000
2018	82,000	82,000 × 60% = 49,200	32,800
2019	72,000	72,000 × 50% = 36,000	36,000
2022	90,000	90,000 × 60% = 54,000	36,000
Total			1,76,600

So, the total number of cars produced by company A which remained unsold in all 6 years together were 1,76,600.

Hence, option (C).

2. (A)

Number of cars produced by company B in 2019 = 40,000

Number of cars that remained unsold by company A in 2015 $= \frac{50}{100} \times 50,000 = 25,000$

Required ratio = 40,000:25,000 = 8:5

Hence, option (A).

3. 8,000

Total number of cars produced by company B in 2015 = 40,000.

Total number of cars sold by company B in 2015 $= \frac{60}{100} \times 40,000 = 24,000$

Total number of cars unsold by company B in 2015 = 40,000 – 24,000 = 16,000.

Total number of cars produced by company B in 2017 = 70,000.

Total number of cars sold by company B in 2017 $= \frac{50}{100} \times 70,000 = 35,000$

Total number of cars unsold by company B in 2017 = 70,000 – 35,000 = 35,000.

Total number of cars produced by company B in 2020 = 65,000.

Total number of cars sold by company B in 2020 $= \frac{50}{100} \times 65,000 = 32,500$

Total number of cars unsold by company B in 2020 = 65,000 – 32,500 = 32,500.

So, total number of cars sold by company B in 2015, 2017, and 2020.

= 24,000 + 35,000 + 32,500 = 91,500

Total number of cars unsold by company B in 2015, 2017, and 2020.

= 16,000 + 35,000 + 32,500 = 83,500

Required difference = 91,500 – 83,500 = 8,000.

Hence, the answer is 8,000.

4. (C)

The number of cars sold by 'A' in 2016 $= \frac{40}{100} \times 58,000 = 23,200$

The number of cars sold by 'B' in 2020

$= \frac{50}{100} \times 65,000 = 32,500$

Required percentage $= \frac{23,200}{32,500} \times 100 = 71.38\%$

Hence, option (C).

5. 12,000

Number of cars sold by company A in 2017 $= \frac{80}{100} \times 60,000 = 48,000$

Number of cars sold by company A in 2019 $= \frac{50}{100} \times 72,000 = 36,000$

Difference $= 48,000 - 36,000 = 12,000$

Hence, the answer is 12,000.

6. (C)

Number of matches played by India = 84

Number of matches won by India = 42

Number of matches lost by India = 42

Number of matches played by New Zealand = 72.

Number of matches won by New Zealand = 51.

Number of matches lost by New Zealand = 21.

Hence, the ratio of the number of matches lost by India and the number of matches lost by New Zealand $= \frac{42}{21} = \frac{2}{1}$

Hence, option (C).

7. (A)

Number of matches played by Sri Lanka = 50.

Number of matches won by Sri Lanka = 10

Winning percentage $= \frac{10}{50} \times 100 = 20\%$.

Number of matches played by South Africa = 60.

Number of matches won by South Africa = 30.

Winning percentage $= \frac{30}{60} \times 100 = 50\%$.

Hence, required difference = (50 – 20)% = 30%

Hence, option (A).

8. 5

Winning percentage of Afghanistan

$= \frac{5}{35} \times 100 = 14.28\%$

Winning percentage of Pakistan

$= \frac{33}{70} \times 100 = 47.14\%$

Winning percentage of Bangladesh

$= \frac{20}{44} \times 100 = 45.45\%$

Winning percentage of Sri Lanka

$= \frac{10}{50} \times 100 = 20\%$

Winning percentage of India

$= \frac{42}{84} \times 100 = 50\%$

Winning percentage of England

$= \frac{45}{80} \times 100 = 56.25\%$

Winning percentage of New Zealand

$= \frac{51}{72} \times 100 = 70.83\%$

Winning percentage of West Indies

$= \frac{14}{35} \times 100 = 40\%$

Winning percentage of South Africa

$= \frac{30}{60} \times 100 = 50\%$

Winning percentage of Australia

$= \frac{50}{90} \times 100 = 55.55\%$

Total five teams are having more than or equal to 50% of the winning ratio.

Hence, the answer is 5.

9. (B)

Winning percentage of South Africa

$= \frac{30}{60} \times 100 = 50\%$

Winning percentage of New Zealand

$= \frac{51}{72} \times 100 = 70.83\%$

Winning percentage of India

$= \frac{42}{84} \times 100 = 50\%$

Winning percentage of Australia

$= \frac{50}{90} \times 100 = 55.55\%$

New Zealand has the highest percentage of wins.

Hence, option (B).

10. (C)

Number of matches played by West Indies = 35.

Number of matches won by West Indies = 14.

Number of matches lost by West Indies = 21.

Hence, the required percentage $= \frac{21}{35} \times 100 = 60\%$.

Hence, option (C).

11. (C)

From point (1) Paul's total marks in all four subjects together = $72.5 \times 4 = 290$.

Therefore, marks obtained by Paul in English = $290 - (80 + 70 + 80) = 60$.

From point (2) total marks in Maths for all four students = $67.5 \times 4 = 270$.

Therefore, marks obtained by Marcus in Maths = $270 - (80 + 60 + 90) = 40$.

So, required total = 60 + 40 = 100

Hence, option (C).

12. (C)

From the explanation of the previous question, we know that Paul scored 60 marks in English and Marcus scored 40 marks in Maths.

Now, from point (3) Bruno's average marks for all four subjects = $1.5 \times 40 = 60$.

Therefore, Bruno's total marks for all four subjects = $60 \times 4 = 240$.

Further, from point (4) Bruno and Edinson scored the same marks in French.

Therefore, Bruno's marks in French = 60.

Bruno's marks in science = 60.

Also, the total marks in maths for all the four students = 270.

So, from point point (5), total marks in science for all the four students = 270 – 10 = 260.

Now, the final table is as follows:

	Maths	Science	English	French	Total
Paul	80	70	60	80	290
Bruno	60	60	60	60	240
Marcus	40	50	60	70	220
Edinson	90	80	70	60	300
Total	270	260	250	270	

From the table, we can say that Marcus scored the lowest total. Hence, option (C).

13. (D)

From the table obtained in the previous question, Edinson scored the highest in Science. Hence, option (D).

14. (C)

From the table obtained in the previous question, Marcus scored the lowest in Maths. Hence, option (C).

15. (A)

From the table obtained in the previous question, Paul scored the highest in French. Hence, option (A).

Level of Difficulty – 2

1. (B)

Revenue = Sales (by volume) × Selling price of each watch = 200 × 100 × 1000 = 2,00,00,000, i.e., ₹2 crore

Profit%

$$= \frac{\text{Profit}}{\text{Expenditure}} \times 100 = \frac{\text{Revenue-Expenditure}}{\text{Expenditure}} \times 100\%$$

$$= \frac{₹2 \text{ Crore-Expenditure}}{\text{Expenditure}} \times 100 = 25\%$$

(as the Profit % is 25%)

So, expenditure = ₹1.6 crore

Hence, option (B).

2. (D)

In the year 2010, the market share of 'Maxima' $= \frac{350}{800} = \frac{7}{16}$

In the year 2011, the market share of 'Maxima' $= \frac{300}{850} = \frac{6}{17}$

In the year 2013, the market share of 'Maxima' $= \frac{300}{950} = \frac{6}{19}$

In the year 2015, the market share of 'Maxima' $= \frac{400}{950} = \frac{8}{19}$

So, we can say that the market share of 'Maxima' is highest in the year 2010. Hence, option (D).

3. (D)

We need the sales revenue and expenditure. As selling prices of watches are not given, so we cannot find sales revenue. We cannot answer the question. Hence, option (D).

4. (B)

By observing the graph, we can say that the sales of Titan in the year 2014 are two times the sales of 2010, but for other watch companies, it is less than double. So, the growth rate is highest for Titan from the year 2010 to 2014. Hence, option (B).

5. (D)

Let's assume that the prices in the year 2016 were ₹20, ₹40, ₹10, ₹30 for Casio, Titan, Maxima, and Noise, respectively.

The following table will help find the answer.

Company	Volume in 2015	Volume in 2016	Price in 2016	Total Revenue
Casio	250	325	₹20	₹6,500
Titan	100	120	₹40	₹4,800
Maxima	400	440	₹10	₹4,400
Noise	200	260	₹30	₹7,800

So, Noise had the highest revenue in the year 2016. Hence, option (D).

6. 2

Average marks of students in batch F = 15

Total marks of seven students in batch F = 15 × 7 = 105.

To pass the test, the most number of students needs to score 40 or above

$\frac{105}{40} = 2.625$

Thus, the lowest integer below it, i.e., maximum of two students can score above cut-off marks, i.e., 40. Hence, the answer is 2.

7. 5

Whenever the average is equal or above the cutoff, all students can qualify (in best-case scenarios). Thus, ACDEG are the possible batches. Hence, the answer is 5.

8. 41

From the previous answers, ACDEG can have seven qualifying students and F can have two qualifying students (best case scenario).

Batch B:

Average marks of students = 26

Total marks of seven students: $26 \times 7 = 182$

To pass the test, the most number of students needs to score 44 or above

$\frac{182}{44} = 4.136$

Thus, the lowest integer below it, i.e., maximum four students can score above cutoff marks, i.e., 44.

So, the highest possible number of students that can qualify = $7 + 7 + 7 + 7 + 7 + 4 + 2 = 41$

Hence, the answer is 41.

9. (B)

To identify the best-performing batch, we need to check how many students can pass in the worst possible scenario.

Let's check for Batch C, as it has a huge difference between the average marks of the batch and the cut-off. (The cut-offs for this batch is pretty low and average score of the students in this batch is high, so chances are high that this will be the best performing batch.)

The average marks of students in Batch C: 63

So, the total marks of seven students: $63 \times 7 = 441$

Also, the maximum marks a student can get is 70.

Consider a case where six students got 70 marks. Adding to a score of 420. So, the 7th student could get $441 - 420 = 21$ marks, which is also above the cutoff marks. In all other cases, where the top scorer marks go down, the bottom scorer marks will go up (as the average marks of the batch are given).

Hence, Batch C is the only batch in which all the seven students would have passed.

So, it will be the best performing batch.

Hence, option (B).

10. 7

Average marks of students in Batch G = 30

Total marks of seven students in Batch G $= 30 \times 7 = 210$

Therefore, the maximum number of students that can pass is seven.

Hence, the answer is 7.

11. (B)

Total number of insects in all the national parks $= 10 + 55 + 60 + 35 + 22 = 182$

Percentage of insects in XYZ national park = 13%

$\therefore$ Number of birds $= \frac{182}{13} \times 15$

$= 14 \times 15$

$= 210$

Hence, option (B).

12. (D)

Total number of animals in insects category

$= 10 + 55 + 60 + 35 + 22$

$= 182 = 13\%$

$\therefore$ Total animals $= \frac{182}{13} \times 100 = 1,400$

Now, animals from DEF national park

$= \frac{182 \times 8}{13}$

$= 112$

Animals from insects category from DEF national parks = 55.

∴ Remaining animals from DEF

$= 112 - 55$

$= 57$

Number of animals who are from bats category

$= \frac{182 \times 10}{13}$

$= 140$

∴ Required percentage $= \frac{57}{140} \times 100 \approx 41\%$

Hence, option (D).

13. (B)

Number of animals from insect category from BCD, CDE, and DEF national

Parks $= 55 + 60 + 35 = 150$

Average $= \frac{150}{3} = 50$

∴ Total number of animals from mammals category from BCD, CDE, and

DEF national parks = 50.

The number of mammals $= \frac{15}{100} \times 1{,}400 = 210$

Now, the number of mammals from ABC and EFG national parks together

$= 210 - 50$

$= 160$

Hence, option (B).

14. (B)

As per our previous calculations,

Total number of animals in XYZ national park is

$= \frac{100}{13} \times 182 = 1{,}400$ animals

Hence, option (B).

15. (D)

As per our previous calculations

Total number of insects imported from five national parks to XYZ national park $= 10 + 55 + 60 + 35 + 22 = 182$, i.e., 13% of total animals in XYZ national park.

So, number of reptiles in XYZ national park $= \frac{25}{13} \times 182 = 350$

Required difference $= 350 - 182 = 168$.

Hence, option (D).

Level of Difficulty – 3

1. (B)

Let us make a table of the percentage breakup of the marks secured by five students in the three sections as shown below:

Students	QA	VARC	LRDI
A	25%	20%	25%
B	10%	25%	10%
C	12%	15%	20%
D	28%	30%	25%
E	25%	10%	20%

It is given that the total marks obtained were the same for exactly three of the five students.

As per their percentage breakup, it can be seen that,

1. Student D has scored way above the mean, i.e., 20% of the total score (100% of total score/5 students = 20%) in all the three sections, and
2. Student A has scored 20% in VARC, but more than the mean in the other two sections.

Hence, the total marks of D and A can never be the same as any one of B, C, or E.

So, we can say that B, C, and E must have scored equal total marks in the three sections.

Now, let us assume the total marks in QA section are 100*x*, the total marks in VARC section are 100*y*, and the total marks in LRDI section are 100*z*.

Then, the marks obtained by students B, C, and E in the three sections will be as shown below:

Students / Sections	B	C	E	Total
QA	10*x*	12*x*	25*x*	100*x*
VARC	25*y*	15*y*	10*y*	100*y*
LRDI	10*z*	20*z*	20*z*	100*z*

Equating the total marks scored by E and C, we get:

$25x + 10y + 20z = 12x + 15y + 20z$

$\Rightarrow 13x = 5y$

$\Rightarrow \frac{x}{y} = \frac{5}{13} \rightarrow \ldots$ (i)

Also, by equating the total marks scored by C and B, we get:

Also, $12x + 15y + 20z = 10x + 25y + 10z$

$\Rightarrow 2x - 10y = -10z$ … (ii)

From equation (i), let $x = 5a$ and $y = 13a$. Putting these values in the equation (ii), we get:

$2(5a) - 10(13a) = -10z$

$\therefore 10a - 130a = -10z$

$\therefore z = 12a$

So, $x = 5a$, $y = 13a$, and $z = 12a$.

Now, our final table showing the marks scored by five students in the three sections will be as shown below:

Students / Sections	A	B	C	D	E	Total
QA	125a	50a	60a	140a	125a	500a
VARC	260a	325a	195a	390a	130a	1,300a
LRDI	300a	120a	240a	300a	240a	1,200a
Total	685a	495a	495a	830a	495a	3,000a

The required percentage for student

$$A = \frac{300a}{685a} \times 100\% = 43.8\%$$

The required percentage for student

$$B = \frac{120a}{495a} \times 100\% = 24.24\%$$

The required percentage for student

$$C = \frac{240a}{495a} \times 100\% = 48.9\%$$

The required percentage for student

$$D = \frac{300a}{830a} \times 100\% = 36.14\%$$

The required percentage for student

$$E = \frac{240a}{495a} = 100\% = 48.9\%$$

Clearly, for three students the marks obtained in LRDI section as a percentage of his/her total marks obtained is greater than 39%.

Hence, option (B).

2. (D)

Let's use the table that we created in the previous question.

Students / Sections	A	B	C	D	E	Total
QA	125a	50a	60a	140a	125a	500a
VARC	260a	325a	195a	390a	130a	1,300a
LRDI	300a	120a	240a	300a	240a	1,200a
Total	685a	495a	495a	830a	495a	3,000a

The marks obtained by D in VARC section = 390a

The marks obtained by A in QA section = 125a

Hence, the required ratio = 390a:125a

= 78:25

Hence, option (D).

3. (C)

From the table that we created earlier,

The marks obtained by C in VARC section = 195a

Also, as per the question, 195a = 292.5

$$\therefore \quad a = \frac{292.5}{195}$$

$$\therefore \quad a = \frac{3}{2}$$

Hence, the marks obtained by D in QA section $= 140a = 140 \times \frac{3}{2} = 210$

Hence, option (C).

4. (A)

From the table that we created earlier,

The marks obtained by E in LRDI section = 240a

$\Rightarrow 240a = 120$

$\Rightarrow a = 0.5$

Hence, the average of the total marks obtained by B

$$= \frac{495a}{3} = \frac{495 \times 0.5}{3} = \frac{247.5}{3} = 82.5$$

Hence, option (A).

5. 0

Let's use the table that we created in the previous question

Students / Sections	A	B	C	D	E	Total
QA	125a	50a	60a	140a	125a	500a
VARC	260a	325a	195a	390a	130a	1300a
LRDI	300a	120a	240a	300a	240a	1200a
Total	685a	495a	495a	830a	495a	3000a

Total marks obtained by C = 495*a*

Total marks obtained by E = 495*a*

Hence, the required difference is 0.

6. (C)

As per the given information, on the 11th, out of 125 eggs, 26% is consumed.

Hence, 26% of 125 = 32.5, i.e., 32 eggs.

Thus, the total number of eggs available on 12th = 236 + (125 – 32) = 329.

So, the number of eggs consumed on the 12th = 71% of 329 = 233.59 = 233, and the total available for the next day will be 423 + (329 – 233) = 519, and so on.

So, we can create the following table for the eggs available and consumed on each date.

Date	Eggs carried from previous day	Eggs bought	Total eggs available in stock	Eggs consumed	Eggs left
11	0	125	125	32	93
12	93	236	329	233	96
13	96	423	519	337	182
14	182	474	656	551	105
15	105	423	528	480	48
16	48	526	574	419	155
17	155	471	626	431	195
18	195	532	727	465	262

So, it is evident from the table that the highest number of eggs available in stock was on the 18th. Hence, option (C).

7. (C)

From the table created in the previous questions, it can clearly be seen that there is a decrease in the number of available eggs only for one day, i.e., 15th (as compared to the 14th). Hence, option (C).

8. (D)

From the table, on the 14th, the consumption of eggs was the highest. Hence, option (D).

9. (B)

Date	Eggs carried from previous day	Eggs bought	Total eggs available in stock	Eggs consumed	Eggs left
11	0	125	125	32	93
12	93	236	329	233	96
13	96	423	519	337	182
14	182	474	656	551	105
15	105	423	528	480	48
16	48	526	574	419	155
17	155	471	626	431	195
18	195	532	727	465	262

From the table, the percentage change in the consumption of eggs on 14th $= \frac{551-337}{337} \times 100 = 63.5\%$.

Hence, option (B).

10. (C)

From the table, the required difference = 727 – 519 = 208. Hence, option (C).

11. (D)

The number of Bluetooth earphones sold by P on Zyntra = 5% of (4,00,000) = 20,000

Similarly, for Q, R, S, and T, the number of Bluetooth earphones sold on Zyntra are 60,000, 1,00,000, 1,40,000 and 80,000, respectively.

The total number of Bluetooth earphones sold by each of these company in India should not be less than the number of Bluetooth earphones sold by the companies on Zyntra.

The total bluetooth earphones sold by P in India is 10% of K. Also, the total Bluetooth earphones sold by P on Zyntra = 5% of 4,00,000 = 20,000

So, 10% of K ≥ 20,000 $\Rightarrow$ K ≥ 2,00,000

Similarly,

For Q, 20% of K ≥ 60,000 $\Rightarrow$ K ≥ 3,00,000

For R, 25% of K ≥ 1,00,000 $\Rightarrow$ K ≥ 4,00,000

For S, 20% of K ≥ 1,40,000 $\Rightarrow$ K ≥ 7,00,000

For T, 25% of K ≥ 80,000 $\Rightarrow$ K ≥ 3,20,000

Thus, the minimum value of K should be 7,00,000

Hence, option (D).

12. (C)

In such questions, instead of calculating the percentage values, we can just find out the fractional values and compare them. The result will be the same.

The fraction of its total Bluetooth earphone sold by P on Zyntra

$= \frac{(5\% \text{ of } 4{,}00{,}000)}{(10\% \text{ of K thousand})} = \frac{200}{K}$

The fraction of its total Bluetooth earphone sold by Q on Zyntra

$= \frac{(15\% \text{ of } 4{,}00{,}000)}{(20\% \text{ of K thousand})} = \frac{300}{K}$

The fraction of its total Bluetooth earphone sold by R on Zyntra

$= \frac{(25\% \text{ of } 4{,}00{,}000)}{(25\% \text{ of K thousand})} = \frac{400}{K}$

The fraction of its total Bluetooth earphone sold by S on Zyntra

$= \frac{(35\% \text{ of } 4{,}00{,}000)}{(20\% \text{ of K thousand})} = \frac{700}{K}$

The fraction of its total Bluetooth earphone sold by T on Zyntra

$= \frac{(20\% \text{ of } 4{,}00{,}000)}{(25\% \text{ of K thousand})} = \frac{320}{K}$

Since P has the minimum fraction of its total sales coming from Zyntra, P has

sold the least percentages of its total Bluetooth earphones through Zyntra. Hence, option (C).

13. (B)

As seen in the previous question, the percentage of its total Bluetooth earphones sold by Q on Zyntra

$= \frac{300}{K} \times 100\%$

The percentage of its total Bluetooth earphones sold by R on Zyntra

$= \frac{400}{K} \times 100\%$

Given, $\frac{300}{K} \times 100\% \geq 25\% \Rightarrow K \leq 12,00,000$

Also, $\frac{400}{K} \times 100\% \leq 50\% \Rightarrow K \geq 8,00,000$

Thus, $8,00,000 \leq K \leq 12,00,000$

So, the maximum value of K (in thousand) = 1,200.

Hence, option (B).

14. (C)

The number of companies with sales equal to or greater than 50% will be maximum when the total sales of these companies in India is minimum, i.e., for K = 7,00,000 (as explained in the previous question).

So, the total number of Bluetooth earphones sold by P = 10% of (700,000) = 70,000.

Similarly, for Q, R, S, and T, the total number of Bluetooth earphones sold are 140,000, 175,000, 140,000, and 175,000 respectively.

We know that the number of Bluetooth earphones sold by P on Zyntra = 5% of (4,00,000) = 20,000. Similarly, for Q, R, S, and T, the number of Bluetooth earphones sold on Zyntra are 60,000, 1,00,000, 1,40,000 and 80,000, respectively.

The percentage of Bluetooth earphones sold on Zyntra by P, Q, R, S, and T are 28.57%, 42.85%, 57.14%, 100%, and 45.71%.

Thus, there are only two companies (R and S) who here sold 50% or more of its total Bluetooth earphones on Zyntra. Hence, option (C).

15. (C)

Number of Bluetooth earphones sold by S in India on Zyntra

= 35% of 4,00,000 = 1,40,000

Number of Bluetooth earphones sold by T in India on Zyntra

= 20% of 400,000 = 80,000

So, the required difference = 1,40,000 – 80,000 = 60,000.

Short-cut

The required difference = (35% – 20%) of 4,00,000 = 60,000

Hence, option (C).

Rack Your Brain

1. Cannot be determined

In 1995, for company A, we have:

$\frac{E_A}{I_A} = 1.75$ ... (i)

(where E_A = amount of exports, I_A = amount of imports of company A is 1995).

In 1995, for company B, we have:

$\frac{E_B}{I_B} = 0.75$ ... (ii)

(where E_B = amount of exports, I_B = amount of imports of company B in 1995].

Also, we have $E_A = 2E_B$ .... (iii)

Since we don't know the exact value of imports in 1995 for company A, the value of imports by company B cannot be determined.

Rack Your Brain

2. $111.6° = 108° + 3.6° = 30\% + 1\%$

So, the answer is 31%.

Rack Your Brain

3. Tabular Data

	1st Year		Second Year		Third Year	
	Male	Female	Male	Female	Male	Female
Science	20	30	30	45	50	75
Commerce	40	60	60	90	100	150
Arts	20	30	30	45	50	75
Total	80	120	120	180	200	300